MIL-G-83172B(USAF)
AMENDMENT 6
19 NOVEMBER 1986
SUPERSEDING
AMENDMENT 5
20 June 1980

MILITARY SPECIFICATION

GYROSCOPE ASSEMBLY, RATE SBU-13/A

This amendment forms a part of MIL-G-83172B(USAF), dated 10 Nov 1971, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Delete "2.1" and add: "2.1 Government documents."

* Add: "2.1.1 Specification standards, and handbooks. The following specifications, standards, and handbooks form a part of this specifcation to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATION, Federal, delete "QQ-P-416" and add:
"PPP-B-601 Boxes, Wood, Cleated Plywood";
"PPP-B-636 Box, Shipping, Fiberboard."

* Under SPECIFICATIONS, Military, delete "MIL-C-4150" and "MIL-P-26514". Add: "MIL-M-7793/6 Meter, Time Totalizing, Miniature, Digital, 115 Volts, 400 Hz and MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited."

* Under STANDARDS, Military, delete "MIL-STD-100" and add: "DOD-STD-100."

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

Under STANDARDS, Military, delete "MS17322".

* Add:
"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:38Z
Check the source to verify that this is the current version before use.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 RAce St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)"

* Add: "3.3.4.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

PAGE 3

3.4.1, Delte and add:
"3.4.1 Overhaul. The gyroscope assembly shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases without the use of special tools unless they are approved by the contracting activity. It shall be so constructed that each gyro will have a separate electronic demodulator. Each gyro and electronic demodulator shall be capable of being repaired/replaced individually with any matching procedures readily made by depot overhaul personnel."

3.4.5 Following last sentence add: "The maintainability parameters shall be in accordance with MIL-STD-471."

PAGE 4

3.5 i Delete "+85°C to -40°C" and add: "+71°C to -57°C".

PAGE 5

3.5.2 Delete entire paragraph

3.5.3 Delete "The gyroscope" and add: "Each gyroscope subassembly."

TABLE I under column entitled "Pitch", delete "0.02°/sec" and add "0.20°/sec."

PAGE 6

* 3.6 Line 3 delete "MIL-STD-100" and add: "DOD-D-100."

PAGE 7

3.10.1 Line 1 delete "in the gyro subassembly case" and add " in the case of each gyro subassembly."

* 3.11 Line 2 delete "MS17322" and add: "MIL-M-7793/6."

PAGE 10

3.17.2 Delete and add:

"3.17.2 Steel Parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:38Z
Check the source to verify that this is the current version before use.

demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

* Add:
"4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

PAGE 14

4.5.2.2e: Delete

PAGE 19

4.6.12 Sensitivity at low temperature. Delete "and the phase linearity shall be + 4° from the best straight line in the band pass."

PAGE 20

4.6.13 Sensitivity at high temperature: Delete " and the phase linearity shall be +4° from the best straight line in the band pass."

PAGE 21

4.6.16 Delete enire paragraph and renumber subsequent paragraphs.

4.6.17 Line 1 delete "The case" and add: "The case of each gyroscope subassembly."

4.6.19 Line 2 delete "method 503" and add: "method 503.1."

4.6.20 Line 2 delete "method 504, class 2" and add: "method 504.1. category 6." Line 3 delete "step 5 and add: "step 6."

4.6.21 Delete entire paragraph.

PAGE 22

4.6.22.1 Delete "method 507" and add "method 507.1."

4.6.22.2 Delete "method 508" and add "Method 508.1."

4.6.22.3 Delete "method 509" and add "method 509.1."

4.6.22.4 Delete "method 510" and add "method 510.1."

4.6.24 Delete "method 516" and add "method 516.2."

4.6.25 Delete and add:



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:38Z
Check the source to verify that this is the current version before use.

"4.6.25 Vibration. The gyroscope assembly shall be subjected to the vibration test specified in MIL-STD-810, method 514.2, procedure IA. During the funtional test, the sensitivity test specified in 4.6.12 shall be performed and met. After the endurance test, the gyroscope shall meet individual tests except 4.6.2, 4.6.8 and 4.6.9."

4.6.26 Delete "method 515" and add: "method 5.15.2."

PAGE 25

Delete "4.6.27.3" and add: "4.6.27.3 Test records. Throughout the reliability test, the contractor shall maintain complete test records and shall keep them available for inspection upon request by the local Government Representative. As a minimum, the records shall include the following in accordance with MIL-STD-781: Operating sheet, log of failure, failure report, equipment repair sheet, failure tag and failed-part summary analysis."

4.6.28 Delete entire paragraph.

4.7 Delete and add:

"4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements. Section 4 of MIL-P-116. The sampling and inspection of the packaging and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First Article. When the unit container is capable of serving as the shipping container FIrst Article Inspection, and rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction."

PAGE 26

Delete Section 5 in its entirety and add:

"5. PACKAGING.

5.1. Preservation-packaging. Preservation-packaging shall be level A, C, or Indsutrial, as specified (see 6.2).

5.1.1. Level A.

5.1.1.1. Cleaning. Gyroscope shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2. Drying. Gyroscope shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:38Z
Check the source to verify that this is the current version before use.

5.1.1.4. Unit Packaging. Unless otherwise specified by the contracting activity, each gyroscope shall be packaged in quantity unit packs of one each in accordance with Method ICI of MIL-C-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values G's in established for the gyroscope when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2. Level C. Each gyroscope shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3. Industrial. The Industrial preservation of gyroscope shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Gyroscope packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required,

5.2.2. Level B. Gyroscope packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3. Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4. Industrial. The packaged gyroscope shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

PAGE 27

Delete "6.3" and "6.3.1."

PAGE 28

Delete "6.5"



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:38Z
Check the source to verify that this is the current version before use.

MIL-G-83172B(USAF)
AMENDMENT 6

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was dones as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
AIR FORCE-99

Preparing Activity:
Air Force-71

Project No:
6615-F134



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:38Z
Check the source to verify that this is the current version before use.